Part A: Conceptual Foundation

Q1. Short Notes

1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, exploring and interpreting data to find useful information, patterns and insights.

The main steps of data analysis are:

1. Collect the data.
2. Clean the data.
3. Explore the data.
4. Analyze patterns and relationships.
5. Interpret the results.
6. Communicate the findings.

In this project, data analysis helps us understand the customer credit risk dataset before applying machine learning techniques.

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2. How to Plan a Data Science Project

A Data Science project can be planned using the following steps:

1. Define the problem.
2. Understand the business objective.
3. Collect the required data.
4. Explore and understand the dataset.
5. Clean the data.
6. Perform feature engineering.
7. Prepare the data for machine learning.
8. Build and evaluate the model.
9. Communicate the results.
10. Deploy the solution if required.

A proper plan helps to complete the project systematically and reduces errors.

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3. How to Frame a Machine Learning Problem

To frame a Machine Learning problem, we first need to clearly define what we want the model to predict.

The main steps are:

1. Define the objective.
2. Identify the target variable.
3. Identify the input features.
4. Decide whether the problem is classification or regression.
5. Select suitable evaluation metrics.
6. Prepare the dataset for modeling.

For this project:

- Objective: Predict whether a customer is likely to default on a loan.
- Target variable: `default_flag`
- `0` = No Default
- `1` = Default
- Problem Type: Binary Classification

Q2. Tensors and NumPy Examples

What is a Tensor?

A tensor is a data structure used to store numerical data in multiple dimensions.

A tensor can be understood as a generalization of scalars, vectors and matrices.

Tensor Dimensions

- 0-D Tensor → Scalar
- 1-D Tensor → Vector
- 2-D Tensor → Matrix
- 3-D Tensor → 3-D Tensor
- Higher-D Tensor → Data with more dimensions

Examples

A scalar contains a single value:

```
5
```

A vector contains a sequence of values:

```
[10, 20, 30]
```

A matrix contains rows and columns:

```
[[1, 2, 3],
 [4, 5, 6]]
```

A 3-D tensor can contain multiple matrices:

```
[
    [[1, 2], [3, 4]],
    [[5, 6], [7, 8]]
]
```